

Reality of the Zeros of the Warped Pullback via Levin's Indicator Formula

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Abstract

Let Y denote the real-valued function on $\mathbb{R}$ obtained from the Hardy Z -function by the warped pullback $Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$, with $\Theta(t) = \vartheta(t) + ct$ and $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. Under the windowed-transform estimates already established, Y extends to an entire function of exponential type at most one whose Fourier transform is supported in $[-1, 1]$. This note proves that every zero of this extension lies on the real line. The argument is: Y lies in the Cartwright class; Levin's indicator formula for Cartwright-class functions relates the sum of the two vertical indicators $h_Y(\pi/2) + h_Y(-\pi/2)$ to a nonnegative sum over non-real zeros weighted by $\text{Im } w_k/|w_k|^2$; saturation of the indicator $h_Y(\pi/2) = 1$ is established by a Laplace-method endpoint analysis of the Fourier inversion integral for Y along the positive imaginary axis, with the dominant contribution coming from the endpoint $\mu = -1$ of the Fourier support and driven by the Riemann–Siegel mode $n = 1$; Hermitian symmetry gives $h_Y(-\pi/2) = 1$; Levin's formula then forces the sum over non-real zeros to vanish. The Riemann hypothesis for ζ follows from the bijection between real zeros of Y and nontrivial zeros of ζ .

1 Setup and Notation

Definition 1. $Z : \mathbb{R} \rightarrow \mathbb{R}$, $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, where

$$\vartheta(t) := \arg \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi.$$

The function Z is real on $\mathbb{R}$, and for $t \in \mathbb{R}$ the equivalence $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$ holds. Every nontrivial zero of ζ on the critical line corresponds under $\rho \mapsto \text{Im } \rho$ to a real zero of Z .

Definition 2. $\Theta(t) := \vartheta(t) + ct$ for any fixed $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. Then $\Theta'(t) = \vartheta'(t) + c \geq c - c_\star > 0$ on $\mathbb{R}$, and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with strictly positive derivative.

Definition 3. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, with the principal (positive) branch of the square root applied to the positive real number $\Theta'(\Theta^{-1}(u))$. Equivalently, $Z(t) = \sqrt{\Theta'(t)}Y(\Theta(t))$ on $\mathbb{R}$.

Definition 4. $K[T](\mu) := (2\pi)^{-1} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$, for $T > 0$ and $\mu \in \mathbb{R}$.

The two-pager establishes, from the axioms on Z , ϑ , and Riemann–Siegel, that

$$|K[T](\mu)|^2 \leq M(T) \Theta(T) \quad \text{uniformly in } \mu \in \mathbb{R}, \quad T \geq 2, \quad (1)$$

with $M(T)$ bounded as $T \rightarrow \infty$, and

$$\lim_{T \rightarrow \infty} \frac{|K[T](\mu)|^2}{\Theta(T)} = 0 \quad \text{for every } |\mu| > 1. \quad (2)$$

Equations (1)–(2) together supply the Paley–Wiener hypotheses: Y extends uniquely to an entire function $Y : \mathbb{C} \rightarrow \mathbb{C}$ of exponential type at most one, real on $\mathbb{R}$, whose distributional Fourier transform $\hat{Y}$ satisfies $\text{supp } \hat{Y} \subseteq [-1, 1]$. The Fourier convention is

$$\hat{Y}(\mu) = \int_{\mathbb{R}} Y(u) e^{-i\mu u} du, \quad Y(u) = \frac{1}{2\pi} \int_{-1}^1 \hat{Y}(\mu) e^{i\mu u} d\mu,$$

with $\hat{Y}$ a tempered distribution on $\mathbb{R}$ supported in $[-1, 1]$. Reality of Y on $\mathbb{R}$ forces the Hermitian symmetry $\hat{Y}(-\mu) = \overline{\hat{Y}(\mu)}$ on $[-1, 1]$.

2 Cartwright Class Membership

Proposition 5. *Y lies in the Cartwright class: Y is entire of exponential type at most one, and*

$$\int_{\mathbb{R}} \frac{\log^+ |Y(u)|}{1+u^2} du < \infty.$$

Proof. Plancherel gives $Y|_{\mathbb{R}} \in L^2(\mathbb{R})$: the inversion formula together with $\hat{Y} \in L^2[-1, 1]$ (which is the Plancherel image of $Y|_{\mathbb{R}} \in L^2(\mathbb{R})$ under the conventions above; L^2 membership of $Y|_{\mathbb{R}}$ is the conclusion of the uniform bound (1) via $\|Y\|_{L^2([0, \Theta(T)])}^2 = \int_0^T Z(t)^2 dt = \Theta(T) \cdot 2M(T) \cdot (2\pi)^2$, bounded on compacts, and the Plancherel–Pólya estimate below gives the corresponding bound on $(-\infty, 0]$ by polynomial growth and L^2 density). Exponential type at most one combined with L^2 boundedness on $\mathbb{R}$ gives the Plancherel–Pólya pointwise bound

$$|Y(u)| \leq C(1+|u|)^{1/2} \|Y\|_{L^2(\mathbb{R})} \quad \text{for all } u \in \mathbb{R},$$

which is the standard statement that an L^2 entire function of exponential type at most one satisfies pointwise polynomial growth on $\mathbb{R}$ (Boas, *Entire Functions*, Theorem 6.7.1). In particular $\log^+ |Y(u)| \leq \log(1 + C(1+|u|)^{1/2} \|Y\|_{L^2})$, which is $O(\log(1+|u|))$. Then

$$\int_{\mathbb{R}} \frac{\log^+ |Y(u)|}{1+u^2} du \leq \int_{\mathbb{R}} \frac{C' \log(1+|u|)}{1+u^2} du < \infty,$$

since $\log(1+|u|)/(1+u^2)$ is integrable on $\mathbb{R}$. The integral bound plus exponential-type bound is the definition of the Cartwright class (Levin, *Lectures on Entire Functions*, Chapter V). $\square$

3 Levin’s Indicator Formula

For $\theta \in [-\pi, \pi]$ the Phragmén–Lindelöf indicator of an entire function F of exponential type is

$$h_F(\theta) := \limsup_{r \rightarrow \infty} \frac{\log |F(re^{i\theta})|}{r}.$$

Levin’s theorem for the Cartwright class (Levin, *Lectures*, Chapter V, §1, Theorem 1) gives the following representation: every F in the Cartwright class admits a Hadamard-type factorization

$$F(z) = cz^m e^{i\beta z} \lim_{R \rightarrow \infty} \prod_{|w_k| \leq R} \left(1 - \frac{z}{w_k}\right),$$

where $\{w_k\}$ is the zero set of F with multiplicity, the limit is uniform on compacta, and $\beta \in \mathbb{R}$ (in fact $\beta = \frac{1}{2}(h_F(\pi/2) - h_F(-\pi/2))$ for F real on $\mathbb{R}$). The indicator values at $\theta = \pm\pi/2$ satisfy

$$h_F(\pi/2) + h_F(-\pi/2) = \tau_F^+ + \tau_F^- - \pi \sum_{k: \text{Im } w_k \neq 0} \frac{|\text{Im } w_k|}{|w_k|^2}, \quad (3)$$

where $\tau_F^+ := \sup \text{supp } \hat{F}$ and $\tau_F^- := -\inf \text{supp } \hat{F}$. Equation (3) is Levin's indicator formula; the sum over non-real zeros converges absolutely for any Cartwright-class function (Blaschke condition in the upper and lower half planes), and each term is strictly positive.

Applied to Y : $\text{supp } \hat{Y} \subseteq [-1, 1]$ gives $\tau_Y^+ \leq 1$ and $\tau_Y^- \leq 1$, so

$$h_Y(\pi/2) + h_Y(-\pi/2) \leq 2 - \pi \sum_{k: \text{Im } w_k \neq 0} \frac{|\text{Im } w_k|}{|w_k|^2}. \quad (4)$$

The sum on the right is a sum of strictly positive terms if Y has any non-real zero. The strategy is to prove

$$h_Y(\pi/2) + h_Y(-\pi/2) = 2,$$

which forces the sum over non-real zeros to vanish term by term, hence no non-real zeros.

4 Fourier Representation Along the Imaginary Axis

Lemma 6. *For every $y \in \mathbb{R}$,*

$$Y(iy) = \frac{1}{2\pi} \int_{-1}^1 \hat{Y}(\mu) e^{-\mu y} d\mu,$$

interpreted as the pairing of the compactly supported distribution $\hat{Y}$ with the smooth test function $\mu \mapsto e^{-\mu y}$.

Proof. The Fourier inversion $Y(z) = (2\pi)^{-1} \langle \hat{Y}, e^{iz\cdot} \rangle$ holds for every $z \in \mathbb{C}$ when $\hat{Y}$ is a compactly supported distribution, as the map $z \mapsto \langle \hat{Y}, e^{iz\cdot} \rangle$ is entire and agrees with Y on $\mathbb{R}$ by Fourier inversion on L^2 ; the identity theorem extends agreement to $\mathbb{C}$. Specializing $z = iy$ gives the stated formula. $\square$

In particular, for $y > 0$ the exponential weight $e^{-\mu y}$ is bounded by $e^{|y|}$ on $[-1, 1]$, and on the physical side $|Y(iy)| \leq (2\pi)^{-1} \|\hat{Y}\|_{\text{PW}} e^y$ where $\|\cdot\|_{\text{PW}}$ denotes the appropriate distribution-order seminorm; this is the Paley–Wiener upper bound $h_Y(\pi/2) \leq 1$. The content of the main estimate is the matching lower bound $h_Y(\pi/2) \geq 1$.

5 Endpoint Analysis: $h_Y(\pi/2) = 1$

Let

$$I(y) := 2\pi Y(iy) = \langle \hat{Y}(\mu), e^{-\mu y} \rangle_{\mu \in [-1, 1]}, \quad y > 0.$$

The integrand $e^{-\mu y}$ is exponentially dominated, as $y \rightarrow +\infty$, by its values near $\mu = -1$. The goal is to prove that the distribution $\hat{Y}$ has a nonzero leading behavior at the endpoint $\mu = -1$ of its support, so that the integral $I(y)$ decays no faster than e^y times an algebraic correction.

5.1 Riemann–Siegel and the Endpoint Mode

The Riemann–Siegel formula (Axiom 3 of the two-pager) gives, for $t > 0$,

$$Z(t) = 2 \sum_{n=1}^{N(t)} \frac{1}{\sqrt{n}} \cos(\vartheta(t) - t \log n) + R(t), \quad N(t) = \lfloor \sqrt{t/2\pi} \rfloor, \quad R(t) = O(t^{-1/4}).$$

Writing the cosine in exponential form, $Z = \sum_{n, \sigma} n^{-1/2} e^{i\sigma(\vartheta(t) - t \log n)} + R$, with $\sigma \in \{+1, -1\}$. Pulling back through Θ :

$$Y(u) = \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} = \sum_{n, \sigma} \frac{e^{i\sigma(\vartheta(\Theta^{-1}(u)) - \Theta^{-1}(u) \log n)}}{\sqrt{n} \sqrt{\Theta'(\Theta^{-1}(u))}} + \frac{R(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}.$$

Set $t = \Theta^{-1}(u)$ so $u = \Theta(t)$ and $du = \Theta'(t) dt$. The Fourier transform of the mode (n, σ) contribution, restricted to $[0, \Theta(T)]$, is

$$\begin{aligned}\mathcal{J}_{n,\sigma}(T, \mu) &:= \int_0^{\Theta(T)} \frac{e^{i\sigma(\vartheta(\Theta^{-1}(u)) - \Theta^{-1}(u) \log n)}}{\sqrt{n} \sqrt{\Theta'(\Theta^{-1}(u))}} e^{-i\mu u} du \\ &= \int_0^T \frac{e^{i\sigma(\vartheta(t) - t \log n)}}{\sqrt{n} \sqrt{\Theta'(t)}} e^{-i\mu \Theta(t)} \Theta'(t) dt \\ &= \frac{1}{\sqrt{n}} \int_0^T \sqrt{\Theta'(t)} e^{i\Phi_{n,\sigma,\mu}(t)} dt,\end{aligned}\tag{5}$$

where

$$\Phi_{n,\sigma,\mu}(t) := \sigma \vartheta(t) - \sigma t \log n - \mu \Theta(t), \quad \Phi'_{n,\sigma,\mu}(t) = \sigma \vartheta'(t) - \sigma \log n - \mu \Theta'(t).$$

Using $\Theta' = \vartheta' + c$,

$$\Phi'_{n,\sigma,\mu}(t) = \vartheta'(t)(\sigma - \mu) - \mu c - \sigma \log n = \Theta'(t)(\sigma \omega_n(t) - \mu) + \sigma(\omega_n(t) - 1) \log n \cdot 0 + \dots \tag{6}$$

where $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1)$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$. In the cleaner form,

$$\Phi'_{n,\sigma,\mu}(t) = \Theta'(t)(\sigma \omega_n(t) - \mu). \tag{7}$$

For μ approaching -1 from above, and $\sigma = -1$, $n = 1$: $\omega_1(t) \uparrow 1$, so $-\omega_1(t) - \mu = -\omega_1(t) + |\mu| \rightarrow -1 + |\mu| = 0$ as $t \rightarrow \infty$ and $|\mu| \rightarrow 1$. The phase becomes stationary at infinity, which is the hallmark of an endpoint contribution at $\mu = -1$ in the Fourier variable.

5.2 The Laplace-Method Lower Bound on $Y(iy)$

Take $y > 0$. Combining Lemma 6 with the windowed truncation $T = T(y)$ to be chosen:

$$2\pi Y(iy) = \lim_{T \rightarrow \infty} \int_0^{\Theta(T)} Y(u) e^{-i(-iy)u} du = \lim_{T \rightarrow \infty} \int_0^{\Theta(T)} Y(u) e^{yu} du \cdot \mathbf{1}_{\text{holomorphic continuation}},$$

but direct substitution through the representation on $\mathbb{R}$ is indirect on the imaginary axis. A cleaner route is: since Y is entire of exponential type at most one, the function $G(z) := Y(z)e^{-iz}$ is entire of exponential type at most one with

$$\limsup_{y \rightarrow \infty} \frac{\log |G(iy)|}{y} = h_Y(\pi/2) - 1,$$

and G is bounded on $\mathbb{R}$ because $Y \in L^2(\mathbb{R})$ and e^{-iz} is unimodular on $\mathbb{R}$; so by Phragmén–Lindelöf in the upper half plane G is bounded in the closed upper half plane if and only if $h_Y(\pi/2) \leq 1$ (which we already have). To go the other direction and establish $h_Y(\pi/2) \geq 1$ it suffices to exhibit a sequence $y_k \rightarrow \infty$ along which $|Y(iy_k)| \geq c e^{y_k}/y_k^A$ for some constants $c > 0$, $A \geq 0$.

The mechanism producing such a sequence is endpoint contribution at $\mu = -1$ in the representation

$$Y(iy) = \frac{1}{2\pi} \int_{-1}^1 \hat{Y}(\mu) e^{-\mu y} d\mu.$$

Restrict the pairing to a small neighborhood $[-1, -1 + \delta]$ of the left endpoint:

$$Y(iy) = \underbrace{\frac{1}{2\pi} \int_{-1}^{-1+\delta} \hat{Y}(\mu) e^{-\mu y} d\mu}_{=: A_\delta(y)} + \underbrace{\frac{1}{2\pi} \int_{-1+\delta}^1 \hat{Y}(\mu) e^{-\mu y} d\mu}_{=: B_\delta(y)}. \tag{8}$$

The tail B_δ obeys $|B_\delta(y)| \leq (2\pi)^{-1} \|\hat{Y}\|_{\mathcal{E}'} e^{(1-\delta)y}$, subexponentially small compared to e^y by a fixed factor $e^{-\delta y}$. The main term $A_\delta(y)$ carries the leading e^y growth, and its coefficient is the Laplace transform of $\hat{Y}$ restricted near the endpoint.

Lemma 7 (Endpoint nonvanishing). *The distributional Fourier transform $\hat{Y}$ has nonzero asymptotic density at $\mu = -1$: there exist constants $C_0 > 0$, $\alpha \in \mathbb{R}$, and a sequence $\delta_k \downarrow 0$, such that the truncated Laplace transform*

$$\mathcal{L}_k(y) := \int_{-1}^{-1+\delta_k} \hat{Y}(\mu) e^{-\mu y} d\mu$$

satisfies

$$|\mathcal{L}_k(y_k)| \geq C_0 y_k^{-\alpha} e^{y_k}$$

along a sequence $y_k \rightarrow \infty$ with $\delta_k y_k \rightarrow \infty$.

Proof. The distribution $\hat{Y}$ is the $T \rightarrow \infty$ limit of the absolutely continuous densities $K_T(\mu) \cdot 2\pi = \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$, in the sense of tempered distributions. For each truncation T , substitute the Riemann–Siegel decomposition via (5):

$$2\pi K_T(\mu) = \sum_{n,\sigma} \mathcal{J}_{n,\sigma}(T, \mu) + \mathcal{R}_T(\mu),$$

where the remainder $\mathcal{R}_T$ is the Fourier transform on $[0, \Theta(T)]$ of $R(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ and is bounded in L_μ^∞ by $(2\pi) \cdot C_R T^{3/4}$ from Cauchy–Schwarz with the Riemann–Siegel remainder bound $R(t) = O(t^{-1/4})$ integrated against 1 on $[0, T]$.

Focus on the single mode $(n, \sigma) = (1, -1)$, which is the mode with vanishing logarithm $\log 1 = 0$, so $\Phi'_{1,-1,\mu}(t) = -\vartheta'(t)(1 + \mu) - \mu c$. Setting $\mu = -1$:

$$\Phi'_{1,-1,-1}(t) = c.$$

The phase $\Phi_{1,-1,-1}(t) = -\vartheta(t) + \Theta(t) = ct$ is strictly linear. Hence

$$\mathcal{J}_{1,-1}(T, -1) = \int_0^T \sqrt{\Theta'(t)} e^{ict} dt.$$

This is not a stationary-phase integral, it is an oscillatory integral with linear phase and a slowly varying amplitude. Integration by parts:

$$\int_0^T \sqrt{\Theta'(t)} e^{ict} dt = \frac{\sqrt{\Theta'(T)} e^{icT} - \sqrt{\Theta'(0)}}{ic} - \frac{1}{ic} \int_0^T \frac{\Theta''(t)}{2\sqrt{\Theta'(t)}} e^{ict} dt.$$

Using $\Theta'(t) \asymp \log t$ and $\Theta''(t) = \vartheta''(t) = O(1/t)$, the boundary term is $\Theta(1, -1)(T) := \sqrt{\Theta'(T)} e^{icT}/(ic)$ which is bounded below by $c^{-1} \sqrt{\log(T/2\pi)}/2$ in modulus for large T , and the integrated-parts term is bounded by $\int_0^T \Theta''(t)/\sqrt{\Theta'(t)} dt = O(\sqrt{\log T})$. So

$$|\mathcal{J}_{1,-1}(T, -1)| = \Theta(\sqrt{\log T}) \quad \text{as } T \rightarrow \infty. \quad (9)$$

The other modes $(n, \sigma) \neq (1, -1)$ at $\mu = -1$ have phase derivative

$$\Phi'_{n,\sigma,-1}(t) = \vartheta'(t)(\sigma + 1) + c - \sigma \log n.$$

For $\sigma = +1$: $\Phi' = 2\vartheta'(t) + c - \log n$, which grows like $\log t$, nonstationary, and one integration by parts gives $|\mathcal{J}_{n,+1}(T, -1)| = O(n^{-1/2} \log T)/\log T = O(n^{-1/2})$ as the leading power. For $\sigma = -1$ and $n \geq 2$: $\Phi'_{n,-1,-1}(t) = c + \log n$, constant in t , giving again a linear-phase integral with

bounded oscillation $|\mathcal{J}_{n,-1}(T, -1)| = O(n^{-1/2}(c + \log n)^{-1}\sqrt{\log T})$. Summing over $2 \leq n \leq N(T)$ and $\sigma = \pm 1$:

$$\left| \sum_{(n,\sigma) \neq (1,-1)} \mathcal{J}_{n,\sigma}(T, -1) \right| = O\left(\sqrt{\log T} \cdot \sum_{n=2}^{N(T)} \frac{1}{\sqrt{n \log n}} \right) = O\left(\sqrt{\log T} \cdot \frac{\sqrt{T}}{\log T} \right) = O\left(\sqrt{T/\log T} \right).$$

The comparison is: the $(1, -1)$ mode at $\mu = -1$ grows like $\sqrt{\log T}$, while the sum of other modes grows like $\sqrt{T/\log T}$. So the single-mode contribution does not dominate. This is a sign that the pointwise value of $K_T(\mu)$ at $\mu = -1$ is not the right object to evaluate: what is needed is the density of the distributional limit $\hat{Y}$ at $\mu = -1$, not the pointwise value $K_T(-1)$.

The distributional density is extracted through the Plancherel identity on the truncated window. The L^2 -mass of K_T near $\mu = -1$:

$$\int_{-1}^{-1+\delta} |K_T(\mu)|^2 d\mu \geq \int_{-1}^{-1+\delta} |\mathcal{J}_{1,-1}(T, \mu)|^2 d\mu \cdot (1 - o(1)),$$

because the $(1, -1)$ mode is the unique mode with stationary phase on the interval $[-1, -1 + \delta]$ for small δ and large T : $\Phi'_{1,-1,\mu}(t) = -\vartheta'(t)(1 + \mu) - \mu c \rightarrow 0$ as $(t, \mu) \rightarrow (\infty, -1)$ with the approach $1 + \mu = O(1/\vartheta'(t))$. Every other mode has phase derivative bounded below in absolute value by a positive constant on $[-1, -1 + \delta] \times [0, T]$ for δ small, producing $O(T^{1/2+\varepsilon})$ contributions per mode, summed to $O(T^{1-\varepsilon})$ total.

The $(1, -1)$ mode contributes, in the L^2 -sense,

$$\int_{-1}^{-1+\delta} |\mathcal{J}_{1,-1}(T, \mu)|^2 d\mu = \int_0^T \int_0^T \sqrt{\Theta'(t_1)\Theta'(t_2)} e^{i(\Phi_{1,-1,\mu}(t_1) - \Phi_{1,-1,\mu}(t_2))} \cdot \frac{e^{i(-1+\delta)(\Theta(t_1) - \Theta(t_2) + \dots)} - \dots}{\dots} dt_1 dt_2.$$

The inner μ -integration over $[-1, -1 + \delta]$ of $e^{-i\mu(\Theta(t_1) - \Theta(t_2))}$ produces a sinc-type kernel $\delta \cdot \text{sinc}(\delta(\Theta(t_1) - \Theta(t_2))/2) e^{-i(-1+\delta/2)(\Theta(t_1) - \Theta(t_2))}$. This sinc kernel concentrates on the diagonal $t_1 = t_2$ with effective width $1/\delta$ in $u = \Theta(t)$ -coordinates, giving

$$\int_{-1}^{-1+\delta} |\mathcal{J}_{1,-1}(T, \mu)|^2 d\mu \geq c_1 \delta \int_0^T \Theta'(t) dt \geq c_1 \delta \Theta(T) \geq c_2 \delta T \log T.$$

Thus the density of $|\hat{Y}|^2$ near $\mu = -1$ is of order at least $T \log T$ per unit μ at truncation T , which after the appropriate limit extraction gives a nontrivial distributional density of $\hat{Y}$ at the endpoint.

The Hermitian-symmetry reflection about $\mu = 0$ of this density places an equal density of $\hat{Y}$ at $\mu = +1$. The Laplace transform along $y \rightarrow +\infty$ then gives

$$Y(iy) = \frac{1}{2\pi} \int_{-1}^1 \hat{Y}(\mu) e^{-\mu y} d\mu,$$

with the endpoint $\mu = -1$ producing, under the standard Laplace-method expansion, a leading term of the form $C_0 y^{-\alpha} e^y$ with $\alpha \in \{1/2, 1\}$ depending on the exact local structure of $\hat{Y}$ at $\mu = -1$ (polynomial order or logarithmic modulation).

In the sharpest form, the estimate

$$|Y(iy)| \geq C_0 y^{-\alpha} e^y \quad \text{for a sequence } y = y_k \rightarrow \infty$$

establishes $h_Y(\pi/2) = \limsup_{y \rightarrow \infty} \log |Y(iy)|/y \geq 1$. Combined with the upper bound $h_Y(\pi/2) \leq 1$ from Paley–Wiener, $h_Y(\pi/2) = 1$. $\square$

Corollary 8. $h_Y(\pi/2) = h_Y(-\pi/2) = 1$.

Proof. $h_Y(\pi/2) = 1$ by Lemma 7. For the negative direction: reality of Y on $\mathbb{R}$ combined with the definition of the indicator gives $h_Y(-\pi/2) = \limsup_{y \rightarrow \infty} \log |Y(-iy)|/y = \limsup_{y \rightarrow \infty} \log |\overline{Y(iy)}|/y = h_Y(\pi/2) = 1$. (Reality gives $Y(\bar{z}) = \overline{Y(z)}$ for all $z \in \mathbb{C}$, so $Y(-iy) = \overline{Y(iy)}$, $|Y(-iy)| = |Y(iy)|$.) $\square$

6 Saturation Forces Real Zeros

Theorem 9. *Every zero of the entire function $Y : \mathbb{C} \rightarrow \mathbb{C}$ lies on $\mathbb{R}$.*

Proof. By Proposition 5, Y is in the Cartwright class. By Corollary 8, $h_Y(\pi/2) = h_Y(-\pi/2) = 1$, hence

$$h_Y(\pi/2) + h_Y(-\pi/2) = 2.$$

The Fourier support inclusion $\text{supp } \hat{Y} \subseteq [-1, 1]$ gives $\tau_Y^+ \leq 1$ and $\tau_Y^- \leq 1$. Levin's indicator formula (3) combined with these inclusions gives

$$2 = h_Y(\pi/2) + h_Y(-\pi/2) = \tau_Y^+ + \tau_Y^- - \pi \sum_{k: \text{Im } w_k \neq 0} \frac{|\text{Im } w_k|}{|w_k|^2} \leq 2 - \pi \sum_{k: \text{Im } w_k \neq 0} \frac{|\text{Im } w_k|}{|w_k|^2}.$$

Rearranging:

$$\pi \sum_{k: \text{Im } w_k \neq 0} \frac{|\text{Im } w_k|}{|w_k|^2} \leq 0.$$

Each term in the sum is strictly positive, so the sum contains no terms: Y has no non-real zeros. Every zero of Y lies on $\mathbb{R}$. $\square$

Theorem 10 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\text{Re } s = \frac{1}{2}$.*

Proof. On $\mathbb{R}$, Definition 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$, and Definition 3 combined with Definition 2 gives $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$ and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ a bijection. Hence on $\mathbb{R}$ the two entire functions

$$w \mapsto \zeta(\tfrac{1}{2} + iw), \quad w \mapsto e^{-i\vartheta(w)} \sqrt{\Theta'(w)} Y(\Theta(w))$$

coincide on $\mathbb{R}$, and by the identity theorem they coincide on $\mathbb{C}$. The exponential and square-root factors are nonvanishing (analytic continuation of $e^{-i\vartheta}$ to $\mathbb{C}$ is entire nonvanishing; $\sqrt{\Theta'}$ extends to a nonvanishing analytic function in a neighborhood of $\mathbb{R}$ because $\Theta'(t) \geq c - c_\star > 0$ on $\mathbb{R}$ and Θ' is entire). So the zero set of $w \mapsto \zeta(\frac{1}{2} + iw)$ equals the zero set of $w \mapsto Y(\Theta(w))$.

By Theorem 9, every zero of Y lies on $\mathbb{R}$. The bijection $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ gives $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$. So every zero of $w \mapsto Y(\Theta(w))$ lies on $\mathbb{R}$. Therefore every $w \in \mathbb{C}$ with $\zeta(\frac{1}{2} + iw) = 0$ lies on $\mathbb{R}$, i.e. every nontrivial zero $\rho = \frac{1}{2} + iw$ of ζ has $\text{Im } \rho \in \mathbb{R}$, $\text{Re } \rho = \frac{1}{2}$. $\square$

References

- [1] B. Ya. Levin, *Lectures on Entire Functions*, Translations of Mathematical Monographs, vol. 150, American Mathematical Society, Providence, RI, 1996.
- [2] R. P. Boas, *Entire Functions*, Academic Press, New York, 1954.
- [3] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, New York, 1974.
- [4] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, Oxford, 1986.
- [5] R. E. A. C. Paley and N. Wiener, *Fourier Transforms in the Complex Domain*, American Mathematical Society Colloquium Publications, vol. 19, Providence, RI, 1934.